

Enhancing Student Support Services for Chinese International Students in Chiang Mai University

Xiaomeng Bai
College of Arts, Media and Technology
Chiangmai University
Chiangmai, Thailand
xiaomeng_bai@cmu.ac.th

Acrapol Nimmolrat
College of Arts, Media and Technology
Chiangmai University
Chiangmai, Thailand
acrapol@camt.info

Abstract—Currently, among the ever-increasing number of international students studying in Thai universities, Chinese students occupy the largest proportion. Due to the particularity of Chinese students, additional support services were required. In addition, targeted student support services will help the institution to better support Chinese students in their study life and enhance the institution's internationalization. Most studies have focused on general support services rather than targeted services. This study explores the current international student support services and additional service requires of educators and Chinese graduate students in College of Arts, Media and Technology in Chiang Mai University. The jobs to be done theory was used in this study. The results of this study show multiple desired outcomes in each job steps that can help university to create tailored student support services according to the distinct demand of educators and Chinese students.

Keywords—international student support service, Chinese students, Jobs-to-be-Done, Chiang Mai University

I. INTRODUCTION

According to the data released by Ministry of High Education, Science, Research and Innovation of Thailand. In the first semester of 2023, the total number of foreign students in Thailand reached 36,060. Among them, there are 21,906 Chinese students, accounting for the highest proportion among Thai universities. Up to 60% of foreign students are from mainland China [1]. Thailand has been becoming a more favorable destination for Chinese students to continue their further study for higher education [2].

Studies show that the most important factors in education quality assurance are: quality of teaching, learning and support service systems for students [3]. The intense competition in international education market [4] and the increase in the number of international students also led to an expanding awareness of the necessity to provide them with targeted and effective support services, while international students always feel their actual study experiences are not matched to their expectations before enrollment [5]. Understanding the specific expectations of international students and provide them with better international student support services can facilitate a university's internationalization improvements, also play an important role in attracting and retaining international students [6] as well as set the stage for future recruitment of international students.

As the top university in the north region of Thailand, Chiang Mai University (CMU) aim at developing students to become the global citizen with virtue, capability as well as social responsibility. As a part of CMU, College of Arts, Media and Technology (CAMT) also share the same goal. Therefore, university and the college provide students with

multiple support services, the services area including financial, mental and physical health, job search, accident welfare, disability student services and sports venue, etc. [7].

However, the common universality services can't meet the specific demand of Chinese graduate students. Also, currently most studies focus on general student support services, with little focus on student support that specific for international students, international students' unique concerns and demand are traditionally overlooked [8]. However, it is necessary for improving the quality of international student support.

This research aiming to investigate existing international student support services of college of arts, media and technology (CAMT) in CMU and suggest additional service requires from educators and Chinese graduate students. This can lead to more understanding of the services that educators and students believe are important to Chinese graduate students, as well as the pain points they found from actual work experiences about the service and the expectations of the service, help the university to improve the student support services. The research utilizes jobs to be done (JTBD) theory, which explains that people use products and services to solve an important task [9]. Following the steps of JTBD, researcher can investigate and integrate data to determine the demand from the perspective of job performers.

II. LITERATURE REVIEW

A. International student support services

According to the perspective of experts, universal international student support mainly including the services that support international students with the area of academic, cultural, and social adaptation during their study life aboard [10]. Academic support includes a variety of services and resources provided by educational institutions to help students achieve academic success, such as education materials, academic tutoring and workshops and other academic-related activities that boost students' learning and development [11]. These support services work collectively to support a holistic academic environment aim to increase student confidence, performance, and overall satisfaction of academic environment. Cultural support aims at providing the support services to help international students to solve the problems of cross-cultural adaptation including language barrier and lack of understanding of the new culture [12][13], such as festival activities that enable universities to assist international students in navigating cultural differences, fostering a sense of belonging and inclusion [14]. The problems mainly involve language barrier and lack of understanding of the new culture [12]. Social adaptation

support involves providing diverse resources to help international students integrate into foreign societies more effectively, addressing their needs for connection, approval, belonging, and security [15]. Generally including integration activities with local students and surrounding community; assistance with visas and other administrative procedures; rental support to help students understand rental agreements, local housing norms, and access to amenities; and career guidance including career planning advice, recruitment information, resume writing and interview skills [16].

Support for international students encompasses services designed to facilitate their academic, cultural, and social adaptation during their studies abroad [17].

B. Chiang Mai University student support service

Student development division of Chiang Mai University develop the website to present student support services that provide to all students study at CMU, the Chinese students can utilize financial support including scholarship program that face to all students and working capital that face to students with financial difficulties, mental and physical health support which assist students cope with health problems, job search service which help career development, provide recruitment information within Thailand, accident welfare that assist students deal with emergency situation, etc. [7]. The website mainly provide information about scope of services and contact details for each support department.

In order to better provide student support services, the university employs educators to assist students in using the services, these educators are the staff members that responsible for providing assistance and resources for students to better utilize support services in their study life, foster student engagement in the university [18].

C. Jobs-to-be-Done

Jobs-to-be-Done (JTBD) theory provide the framework that help identifying exactly what job performers need, allowing company to provide more effective products or services [19]. Christensen proposed the “job” is what people really strive to accomplish in a circumstance [20]. A job can be divided into functional social and emotional tasks. Functional jobs define the specific tasks that the job performer try to accomplish. Emotional jobs reflect what they desire to avoid during or after completing the functional job. The framework helps to identify the desired outcomes and pain points. Desired outcomes are indicators of success in achieving the job's goal. Pain points are the challenges or obstacles during the job execution [21]. The ultimate goal of a customer using a product or service is to successfully complete the “job” they have, and JTBD is a method to explore customer needs and motivations, providing a reliable foundation for customer-centric service development [22].

JTBD framework request all processes begin with a core functional job-to-be-done, and then the steps taken to achieve that job (Define, Locate, Prepare, Confirm, Execute, Monitor, Modify and Conclude) [22]. In between those steps, researcher can find many desired outcomes that the user has in addition to the main functional job and create job map [23]. According to Jim Kalbach [24], the JTBD process for creating job map including scope the job performer, identify their target job and do the investigation to identify the need

of job performer, the job performer is the stakeholder related in a certain task which have several related jobs, the target job should be the main job that need to be addressed urgently.

Ulwick argues that a job map does not show what the customer is doing, it is not about the solution; rather, it describes what the customer is trying to get done. The format of job statement is structured as “Job statement = verb + object of the verb (noun) + contextual clarifier” [22]. Bettencourt proposed the desired outcome statement format of JTBD. The desired outcomes for direction of improvement in the statement is limited to minimize or maximize [25].

III. METHODOLOGY

This study aiming to investigate existing international student support services of college of arts, media and technology (CAMT) in CMU and suggest additional service requires of educators and Chinese graduate students based on JTBD theory. To systematically discover the jobs that educators and students need to get done, this research was carried out in four parts. Firstly, a literature review was conducted to understand the general international student support service and current student support services in CMU for identifying the job performers and their target jobs. Secondly, adopt semi-structured interview to understand additional service requires of educators and students. Thirdly, create job maps of the target job. Finally, verify the results with the respondents.

A. Data collection

Data collection took place at CAMT of CMU. For collecting details of respondents' information, the semi-structured interview was used to collect data from the four educators of international student service unit and graduate student department and ten second-year graduate Chinese students that study in CAMT. There are total five educators in the international student service unit and graduate student department, considering the schedule of educators, the researcher invited four of them to take the interview. These educators have in-depth understanding and directly experiences of providing international student support services to Chinese students, enable them to provide detailed information. Among CMU, CAMT has the largest number of Chinese students, which can be suitable for sample collecting. Moreover, second-year graduate Chinese students in the knowledge and innovation management (KIM) major with total eighteen students are the group of students just be in CMU for less than two years which can represent the fresh experience of student support service, considering the schedule of students, the researcher invited ten of them to take the interview, this group of respondents can identify new requirements based on their experiences.

Considering practical situations such as respondents' locations and time availability, interviews will be conducted both online and onsite. The interview for each respondent is expected to last approximately 0.5-1 hour.

B. Data analysis

After conducting the interviews, the researcher organized interview records and use the JTBD analysis method to analyse the notes from the individual interviews related to the tasks that educators and students need to get done. The

analysis used the universal job map from Ulwick by mapping the interview findings into the eight job steps, define one core functional job and several detailed functional jobs for each job step. Job definition sentences follow the job statement format presented by Ulwick. The desired outcome statement sentences are formulated according to Job definition. In order to summarize the demand of improvement more clearly, follow the format presented by Bettencourt, the direction of improvement in the statement is limited to minimize or maximize. This analysis allowed for a more structured conclusion of the collected data. Following this, desired outcome job maps were developed to conclude the tasks encountered by educators and students who provide and utilize student support services in daily life.

IV. RESULT

The jobs of educators and Chinese graduate students are crucial for student support services in CAMT, CMU. The results will be shown from two aspects, including students' and educators' perception. For each job step, a set of desired outcomes were collected.

A. Results of the student interview

The desired outcomes of Chinese graduate students were defined for the job steps in the targeted job "Utilize student support service to improve study life quality". Using the universal job map presented by Ulwick, the semi-structured question pattern was built according to the eight steps each job includes. Based on the job steps a main question flow was created as shown in Fig.1.

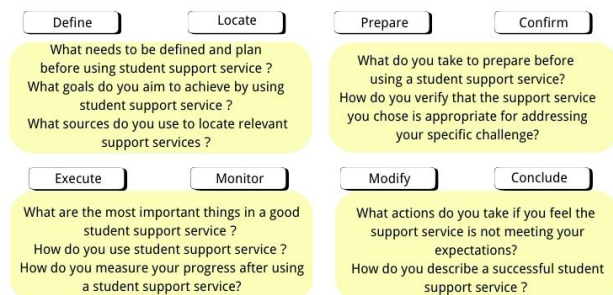


Fig. 1. JTBD interview plan in universal job map for students

The main part of the student interview was broken into three parts: preparation for using student support services (define, locate, prepare), using student support services (confirm, execute) and monitor the success of the student support service (monitor, modify, conclude). The core functional jobs and additional functional jobs of each job step is presented in the Table I below.

TABLE I. FUNCTIONAL JOBS OF STUDENTS

Steps	Core functional jobs	Additional functional jobs
Define	Clarify service demand in study life	Resolve visa-related issues efficiently. Access guidance for academic challenges. Manage stress or depression. Overcome language barriers. Find suitable jobs after graduate.

		Foster connections with peers and understanding of local culture.
Locate	Locate information of required services or resources	Identify the range of student support services available. Receive prompt and effective replies when contacting educators. Use online services with clear, centralized guideline and navigation tools. Find recruitment information targeted at Chinese students.
prepare	Prepare materials or information for using services	Prepare documents that meet official requirements (visa apply, academic program). Access up-to-date transcripts to complete visa-related processes. Sign up for events or activities within the registration period.
Confirm	Confirm service details	Confirm schedule of academic programs and cultural activities. Confirm the accuracy and completeness of prepared documents. Possess accurate information and process of services.
Execute	Start service usage process	Overcome the language barrier contacting with service providers. Gain effective information and solutions for current problems Complete course enrollment without delays or technical issues. Gain opportunities to communicate with students from other countries during social activities.
Monitor	Monitor the effectiveness of the services	Ensure the service content matches personal requirements. Observe improvements in personal development indicators (academic attainment, range of social activities, cultural understanding) Reach service providers easily after receiving initial assistance.
Modify	Overcome obstacles encountered	Get additional help from educators. Find further information from alternative resources. Collect more data for research.
Conclude	Finish service process	Summarize information after complete services session. Reduce confuse of the accuracy of the received advice.

In table I, the target job was divided into eight core functional jobs of each step. From "Clarify service demand in study life" to "Finish service process", each core functional job contains two to five additional functional jobs. This table shows exactly what kind of jobs students are expected to complete in each step when they utilize student support services.

Following Bettencourt's desired outcome statement format, the desired outcomes for direction of improvement in the statement of Chinese graduate students are presented in the Table II below.

TABLE II. DESIRED OUTCOMES OF STUDENTS

Steps	Desired outcomes
Define	Maximize ability of overcoming visa problems, academic challenges and language barriers.
	Minimize stress and depression caused by study life.
	Maximize understanding of local culture.
	Maximize ability to find satisfied jobs.
	Maximize opportunities to broaden social circle.
Locate	Maximize awareness of the range of student support services available.
	Maximize ease of navigating online services.
	Minimize the time cost of communicating with educators.
	Maximize the availability of recruitment information specifically targeted at Chinese students.
Prepare	Minimize errors in preparing documents that meet official requirements for visa and academic program.
	Maximize access of up-to-date transcripts for visa-related processes.
	Maximize opportunities to sign up for activities.
Confirm	Maximize accuracy of the schedule of academic programs and cultural activities
	Maximize ability to confirm the accuracy and completeness of prepared documents.
	Maximize understanding of accurate information and processes of services.
Execute	Minimize language barriers contacting with educators.
	Minimize obstacles of course enrollment.
	Maximize understanding of the effective information for current problems
	Maximize opportunities to communicate with students from other countries during social activities.
Monitor	Maximize the alignment between service content and personal requirements.
	Maximize observable improvements in personal development indicators of academic attainment, range of social activities and cultural understanding.
	Minimize the obstacle of reaching service providers after receiving initial assistance.
Modify	Maximize the availability of additional help from staff members.
	Maximize access of further information from alternative resources.
	Maximize ability to collect data individually.
Conclude	Maximize ability to summarize information after receiving services.
	Minimize confuse of accuracy of the received advice.

In table II, the clear direction of improvement for each functional job was concluded. 27 desired outcomes were defined for the job steps in “Utilize student support service to improve study life quality” job.

Regarding all job steps, the indicators Chinese graduate students concern mainly include ability, obstacles, information accessibility and accuracy, educators engagement and language barrier, etc. The support service areas students mainly used include academic support, cultural support, social activities and visa assistance. Additional support require areas are mental support, career developing support and language support.

B. Results of the educator interview

The desired outcomes of Chinese students were defined for the job steps in the core functional job “Create student

centric student support services”. Based on the job steps a main question flow was created below.

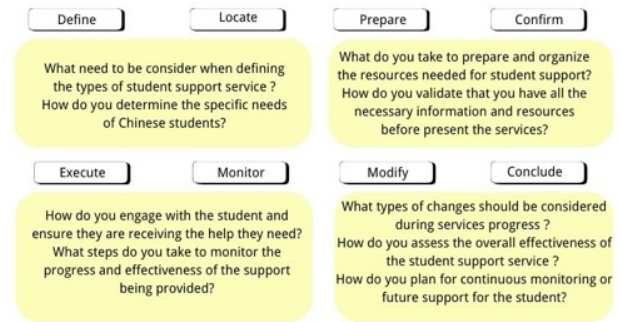


Fig. 2. JTBD interview plan in universal job map for educators

The main part of the educator interview was broken into three parts: preparation for creating student support services (define, locate, prepare), execute student support services (confirm, execute) and monitor the success of the student support service (monitor, modify, conclude). The core functional jobs and additional functional jobs of each job step is presented in the Table III below.

TABLE III. FUNCTIONAL JOBS OF EDUCATORS

Steps	Core functional jobs	Additional functional jobs
Define	Identify student support services types	Identify the specific needs of Chinese students.
		Allocate the budget to design feasible and effective support services.
		Prepare sufficient materials to support students effectively.
Locate	Locate students' needs	Gather opinions from Chinese students about the provided activities and services by surveys and questionnaires.
		Use evaluation form to collect and organize students' feedback.
		Increase Chinese students' response rate for collecting feedback.
prepare	Organize resources for services	Prepare the up-to-date information of services.
		Conduct meetings to evaluate current Chinese student support services.
		Expand services to meet the needs of Chinese students.
		Cooperate with other departments to get the resources of services.
Confirm	Confirm service availability	Verify the accuracy and completeness of service information and documents.
		Recheck the schedules and details of the services.
		Coordinate with students to confirm the meeting time and location.
Execute	Start service process	Provide Chinese students with comprehensive description of the service process.
		Actively engage students in the service process.
		Send emails to answer student questions and notify them about the service process.
		Send required documents to students.
		Address cultural and language barriers to avoid misunderstandings.

		Provide contact information for staffs responsible for specific services.
Monitor	Follow up service completion progress	Collect feedback from Chinese students about the support they received.
		Track students' progress of the services they use.
		Monitor student performance indicators, such as GPA and research progress.
Modify	Solve problems in service process	Offer alternative solutions to students' problems.
		Improve the efficiency of giving more advice to students
		Collect more data for research.
Conclude	End service process	Evaluate effects and deficiencies of the current services.
		Summarize feedback and discuss results for future service improvements.
		Inform Chinese students about services improvements and newly developed services.

In table III, the target job was divided into eight core functional jobs of each step. From “Identify student support services types” to “End service process”, each core functional job contains two to six additional functional jobs. This table shows exactly what kind of jobs educators are expected to complete in each step when they provide student support services to Chinese students.

Following Bettencourt's desired outcome statement format, the desired outcomes for direction of improvement in the statement of educators are presented in the Table IV below.

TABLE IV. DESIRED OUTCOMES OF EDUCATORS

Steps	Desired outcomes
Define	Maximize understanding of Chinese students' needs.
	Maximize the allocation of budget to design feasible and effective support services.
	Maximize the preparation of sufficient materials to support students effectively.
Locate	Minimize the time cost of gathering opinions from Chinese students about provided activities and services.
	Maximize the practicability of evaluation forms to collect and organize students' feedback.
	Maximize the response rate of Chinese students for collecting feedback.
Prepare	Maximize update frequency of service information.
	Maximize the effectiveness of meetings to evaluate current Chinese student support services.
	Maximize the coverage of services to meet the needs of Chinese students.
	Maximize the cooperation with other departments to acquire service resources.
Confirm	Maximize the accuracy and completeness of service information and documents.
	Minimize errors of schedules and details of services.
	Maximize coordination with students to confirm meeting times and locations.
Execute	Maximize the comprehensiveness of descriptions about the service process provided to Chinese students.
	Maximize student engagement in the service process.
	Maximize efficiency of sending emails to answer student questions and notify them about the service process.

	Minimize misunderstandings caused by cultural and language barriers.
	Maximize the accessibility of contact information for staffs responsible for specific services.
Monitor	Maximize the feedback collected from Chinese students about the support they received.
	Maximize understanding of students' progress of using the services.
	Maximize attention for students' performance change.
Modify	Maximize ability to provide alternative solutions for students' problems.
	Maximize the efficiency of giving advice to students.
	Maximize ability to collect extra data.
Conclude	Maximize the evaluation of effects and deficiencies of current services.
	Maximize the summarization of feedback and discussion results for future service improvements.
	Maximize the timeliness of informing Chinese students about services improvements and newly developed services.

In table IV, the clear direction of improvement for each functional job was concluded. 27 desired outcomes were defined for the job steps in “Create student centric support services” job.

Regarding all job steps, the indicators that educators most concern are student engagement and feedback, language barrier, information timeliness, clarity and certainty, service efficiency and certainty and department cooperation, etc. According to the educators, the international student support services in CAMT currently including the areas of academic support, cultural support such as festival activities, social adaptation support such as visa assistance, mental health support, orientation activities and recruitment fair. The educators planning to improve the Chinese student support services, in addition to enriching the existing service areas, native language support to facilitate communication, rental information services and career development guidance services are considered as well, aiming to provide Chinese students with a better study life experience.

V. CONCLUSION AND DISCUSSION

The research indicate that both students and educators have certain expectations for the improvement of Chinese student support services, considering the services can be upgraded to provide beneficial impact on students' study life as well as on the efficiency of educators' workflows. Both groups of respondents identified shortcomings in the provision and utilization of these services and expressed corresponding needs and expectations for improvement, their shared expectations mainly include overcome language barrier, improve the quality and efficiency of support services, enhance engagement between students and educators.

According to the results, the researcher's suggestions are as follow. For academic support, enhance the connections between students and educators, upgrade the course enrollment website to reduce system failures, shorten maintenance time, and ensure availability; develop academic program for enhancing students' ability to collect and organize data for academic researches. For cultural support, provide language program to assist Chinese students overcome language barriers; also, more cultural activities should be organized to enable students have more understanding of local culture. For social adaptation support, provide students with more solutions for visa problems,

ensure the accuracy and timeliness of documents; organize social activities enable Chinese students to communicate with local students; provide bilingual mental health counseling services and job recruitment information specific for Chinese students. For overall support services, provide Chinese students with support services platform, including the up-to-date information of all the services for Chinese students; create dedicate communication channels that allow educators and Chinese students communicate and solve problems without delay; utilize service evaluation tools to collect and analyze students' feedbacks after service process.

Compared with universal international student support services [10] and existing student support services in CAMT, CMU. The most significant additional requirements of the respondents in each service areas are as follow: academic programs, tutoring, activities and course enrollment of academic support; language aid of cultural support; visa assistance, social activities and job recruitment information of social adaptation support.

Through JTBD analysis, the researcher is able to dig deeper into the main jobs that students and educators need to get done, the researcher found that Chinese graduate students pay more attention to whether student support services can help with improving academic ability, overcoming the obstacles of language barriers and visa issues. These are the areas students mentioned the most during the interview. The main reason of these perspectives is students came to Thailand for further study to master more knowledges and skills, at the same time, the problems they currently face are mainly caused by ineffective communication and complicated visa apply process. Meanwhile, the educators expect to improve work efficiency and effectiveness, through simplify work flows with minimal waste of time and resources, focus on the desire outcomes of students, ensuring the services provided align with students' demand, to better assist Chinese students in completing their study life.

This study provides valuable insights and specific directions for College of Arts, Media and Technology in Chiang Mai University to improve its services for supporting Chinese international students. Future research should expand the sample size and consider including students of different background, major, and age in the research scope to obtain more comprehensive demands, also collect quantitative data and using statistical analysis to help identify the main scope and pattern of services that need to be enhanced. Improving the universality of research conclusions, and provide more suggestions for Chinese student support services improvement.

REFERENCES

- [1] Ministry of High Education, Science, Research and Innovation of Thailand. (2023, February 28). Annual report 2023. <https://www.mhesi.go.th/index.php/all-media/book-ministry/10022-2566-e-book.html>.
- [2] Yin, H., Ruangkanjanases, A., & Chen, C. (2015). Factors affecting Chinese students' decision making toward Thai universities. *International Journal of Information and Education Technology*, 5(3), 189.
- [3] Hill, Y., Lomas, L., MacGregor, J. (2003). Students' perceptions of quality in higher education. *Quality Assurance in Education*, Vol. 11 No. 1, pp. 15-20.
- [4] Songsathaphorn, P., Chen, C., & Ruangkanjanases, A. (2014). A study of factors influencing Chinese students' satisfaction toward Thai universities. *Journal of Economics, Business and Management*, 2(2), 105-111.
- [5] Kelo, Maria, Timothy Rogers, and Laura E. Rumbley. 2010. International student support in European higher education: Needs, solutions, and challenges. Bonn: Lemmens Medien.
- [6] Choudaha, Rahul, and Paul Schulmann. 2014. Bridging the gap: Recruitment and retention to improve international student experiences. NAFSA.org. www.nafsa.org/epubs. Accessed 1 Dec 2016.
- [7] Student Development Division of Chiang Mai University. (n.d.). Chiang Mai University. <https://sdd.oop.cmu.ac.th>
- [8] Mori, Sakurako Chako. 2000. Addressing the mental health concerns of international students. *Journal of Counseling & Development* 78 (2): 137-144.
- [9] Christensen, C M., Hall, T., Dillon, K., Duncan, D. (2016). Competing against luck: The story of innovation and customer choice. Harper Collins.
- [10] Ammigan, R., & Perez-Encinas, A. (2020). International student services. In *The International Encyclopedia of Higher Education Systems and Institutions* (pp. 1860-1863). Dordrecht: Springer Netherlands.
- [11] Collins, W., & Sims, B. C. (2013). Help seeking in higher education academic support services. In *Help seeking in academic settings* (pp. 203-223). Routledge.
- [12] Chen, P., You, X., & Chen, D. (2018). Mental Health and Cross-Cultural Adaptation of Chinese International College Students in a Thai University. *International Journal of Higher Education*, 7(4), 133-142.
- [13] Rujiprak, V. (2016). Cultural and Psychological Adjustment of International Students in Thailand. *International Journal of Behavioral Science*, 11(2).
- [14] McKinlay, N. J., Pattison, H. M., & Gross, H. (1996). An exploratory investigation of the effects of a cultural orientation programme on the psychological well-being of international university students. *Higher Education*, 31(3), 379-395.
- [15] Westwood, M. J., & Barker, M. (1990). Academic achievement and social adaptation among international students: A comparison groups study of the peer-pairing program. *International journal of intercultural relations*, 14(2), 251-263.
- [16] Kelo, M., Roberts, T., & Rumbley, L. (2010). International student support in European higher education: Needs, solutions and challenges. Bonn, Germany: Lemmens Medien.
- [17] Ammigan, R., & Perez-Encinas, A. (2020). International student services. In *The International Encyclopedia of Higher Education Systems and Institutions* (pp. 1860-1863). Dordrecht: Springer Netherlands.
- [18] Owusu-Agyeman, Y., & Moroeroe, E. (2022). Professional community and student engagement in higher education: rethinking the contributions of professional staff. *Journal of Professional Capital and Community*, 7(2), 126-143.
- [19] Whitaker, H. J., Ghebremichael-Weldeselassie, Y., Douglas, I. J., Smeeth, L., & Farrington, C. P. (2018). Investigating the assumptions of the self-controlled case series method. *Statistics in medicine*, 37(4), 643-658.
- [20] Christensen, C.M, Hall, T., Dillon, K., and Duncan, D.S. (2016) *Know your customers' jobs to be done*. Harvard business review 94, 9, 54-62.
- [21] Penkert, L. C., & Pretorius, M. C. (2024, March). Applying the Jobs-to-be-done Framework to understand Hoteliers' Needs on Accommodation Platforms. In *International Conference on Tourism Research* (Vol. 7, No. 1, pp. 331-339).
- [22] Ulwick, A. W., & Osterwalder, A. (2016). Jobs to be done: theory to practice. (*No Title*).
- [23] Vélez, V. J. S. (2021). Enhancing human-centered design methods through jobs to be done (Doctoral dissertation).
- [24] Kalbach, J. (2020). The jobs to be done playbook: Align your markets, organization, and strategy around customer needs. Rosenfeld Media.
- [25] Bettencourt, L. A. (2010). Service innovation: How to go from customer needs to breakthrough services. (*No Title*).